

EXAM #1

Answer following questions. Solve the problems on engineering paper. You must show **all your works** clearly.

1. (10 pt) The voltage and current at the terminals of an automobile battery during a charge cycle are shown in Figure 1.

- Calculate the total charge transferred to the battery.
- Calculate the total energy transferred to the battery.

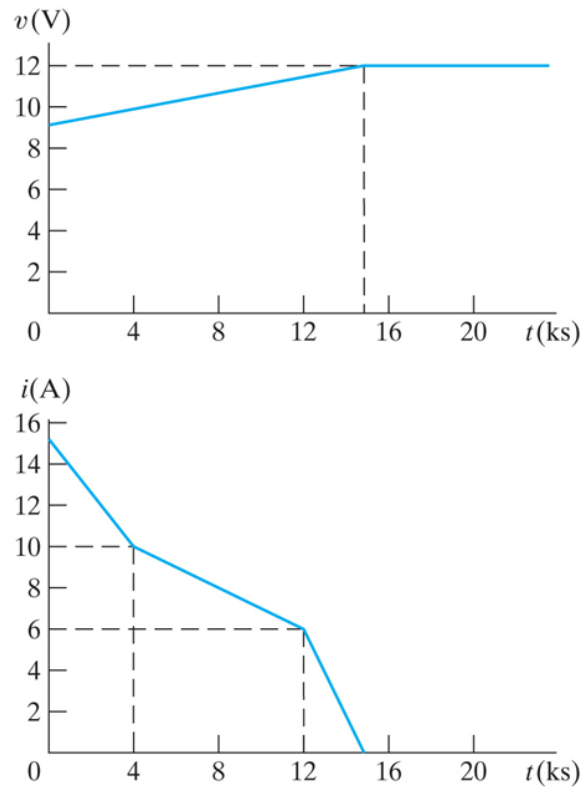


Figure 1: Problem 1

2. (20 pt) For the circuit shown Figure 2 find (a) the current i_i , (b) the voltage v , (c) the total power generated, and (d) the total power absorbed.

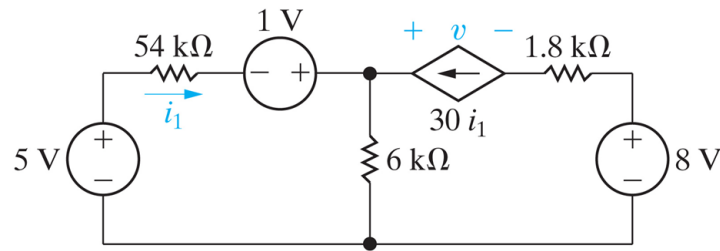


Figure 2: Problem 2

3. (15 pt) The variable resistor R in the circuit in Figure 3 is adjusted until v_a equals 60V. Find the value of R .

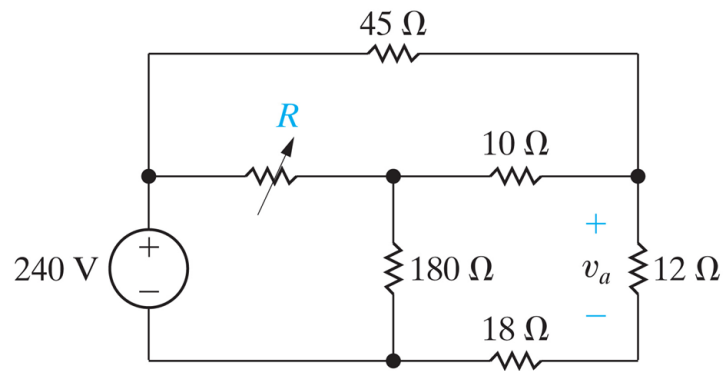


Figure 3: Problem 3

4. (15 pt) The voltage across the $15k\Omega$ resistor in the circuit in Figure 4 is 500V, positive at the upper terminal. Find the power dissipated in each resistor.

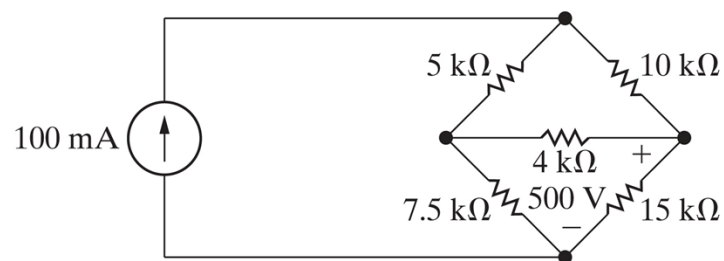


Figure 4: Problem 4